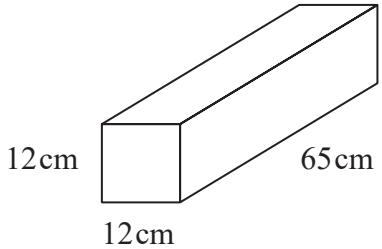
25 The diagram shows a concrete block on horizontal ground.



$$p = \frac{F}{A}$$

p = pressure
F = force
A = area

The block is a cuboid, 12 cm by 12 cm by 65 cm.
The block exerts a force of 220 newtons on the ground.

Calculate the pressure that the block exerts on the ground.

Give your answer in newtons / cm^2

Contact area = 12 × 65 = 780 cm^2

1 mark
Use the face on the ground.

$p = F \div A = 220 \div 780$

$p = 0.28205...\ N/cm^2 \approx 0.282\ N/cm^2$

1 mark
0.282 N/cm^2.

.....newtons / cm^2

(Total for Question 25 is 2 marks)